

The USAGE clause

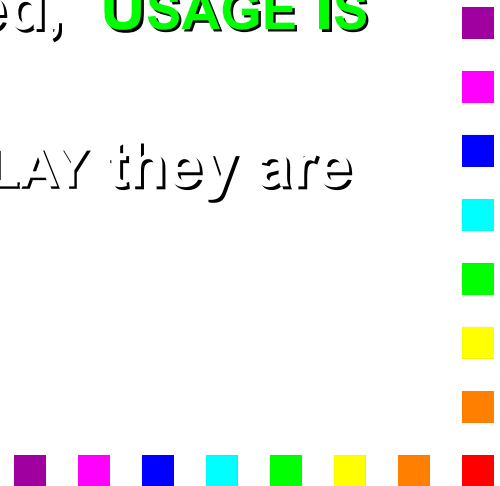
USAGE IS DISPLAY

SYSTEM	CHAR	HEX	DEC	8	4	2	1	8	4	2	1
ASCII	"A"	41	65	0	1	0	0	0	0	0	1
EBCDIC	"A"	C1	193	1	1	0	0	0	0	0	1

The USAGE clause specifies the **format** of **numeric data items** in the computer storage.

When no USAGE clause is explicitly declared, **USAGE IS DISPLAY** is assumed.

When numeric items have a USAGE of DISPLAY they are held as **ASCII** characters !!



Arithmetic when USAGE IS DISPLAY

[illegible]

ADD Num1,Num2 GIVING Num3.

Arithmetic when USAGE IS DISPLAY

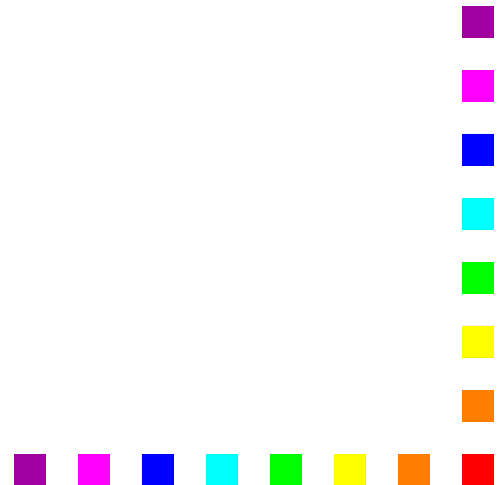
		8	4	2	1	8	4	2	1	HEX	CHAR
Num1	PIC 9 VALUE 4.	0	0	1	1	0	1	0	0	41	"4"
Num2	PIC 9 VALUE 1.	0	0	1	1	0	0	0	1	31	"1"
Num3	PIC 9.	0	1	1	0	0	1	0	1	65	"e"

ADD Num1,Num2 GIVING Num3.

Why use the USAGE clause?

The **USAGE** clause is used for purposes of optimisation - both speed and storage.

The **USAGE** clause controls the way data items (normally numeric) are stored in memory.



USAGE Syntax

[USAGE IS] {
 {
 BINARY
 COMPUTATIONAL
 COMP
 INDEX
 PACKED-DECIMAL
 DISPLAY
 }

01 Num1 PIC 9(5)V99 USAGE IS COMP.

01 Num2 PIC 99 USAGE IS PACKED-DECIMAL.

01 IdxItem USAGE IS INDEX.

01 GroupItems COMP.

02 Item1 PIC 999.

02 Item2 PIC 9(4)V99.

02 New1 PIC S9(5) COMP SYNC.



USAGE IS COMP

COMP items are held in memory as pure binary two's complement numbers.

Number of Digits	Storage Required.
1 TO 4	1 WORD (2 Bytes)
5 TO 9	1 LONGWORD (4 Bytes)
10 TO 18	1 QUADWORD (8 Bytes)

01 TotalCount PIC 9(7) USAGE IS COMP.

Takes 4 bytes (1 LONGWORD) of storage.

COMP - Storage Requirements

2 Bytes can represent any 4 digit number.

32768 16384 8192 4096 2048 1024 512 256 128 64 32 16 8 4 2 1

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Max. number = + or - 32767
PIC 9999

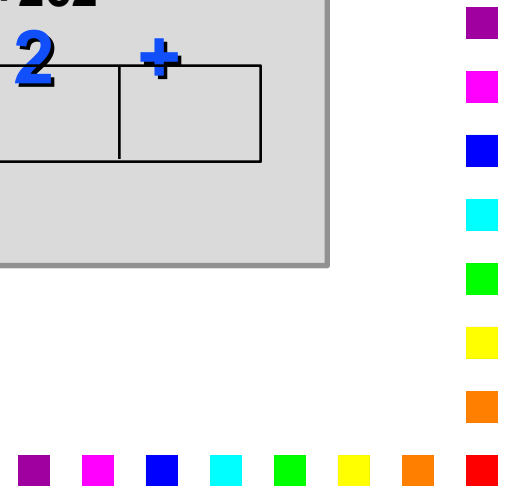
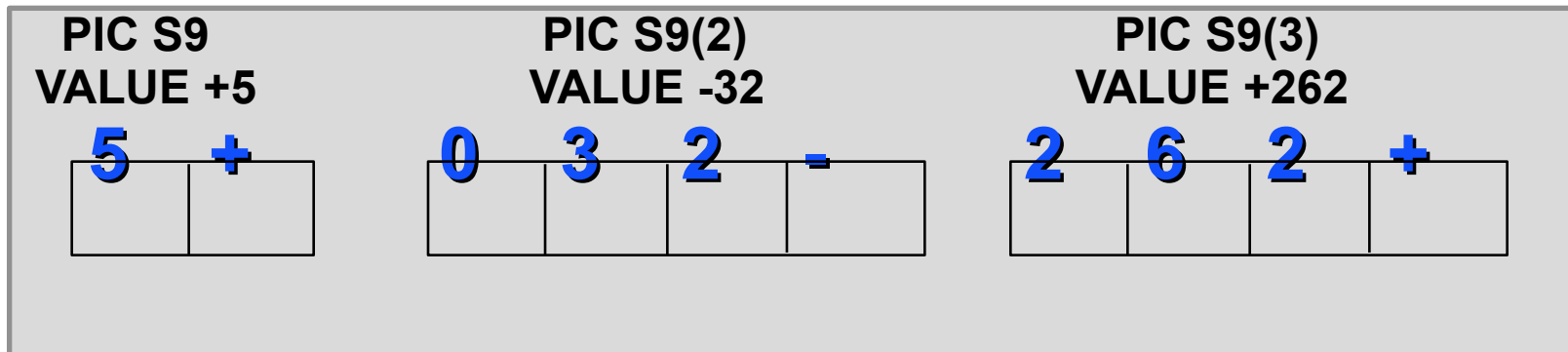
4 Bytes can represent any 9 digit number

Max. number = + or - 2, 147,483,647
PIC 999 999 999

USAGE IS PACKED-DECIMAL

Data items declared as PACKED-DECIMAL are held in **binary-coded-decimal** (BCD) form.

Instead of representing the value as one single binary number, each digit is held in binary form in a nibble (half a byte).



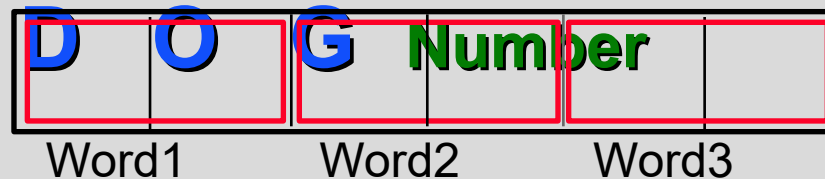
The SYNCHRONIZED Clause

The SYNCHRONIZED clause explicitly aligns COMP and INDEX data items along their natural WORD boundaries.

If there is no SYNCHRONIZED clause the data items are aligned on byte boundaries.

This clause is used to optimise speed of processing - but it does so at the expense of storage.

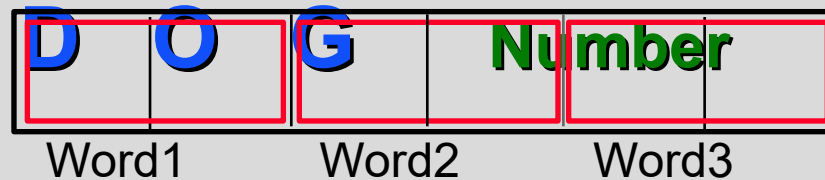
```
01 ThreeBytes    PIC XXX VALUE "DOG".  
01 TwoBytes     PIC 9(4) COMP.
```



The SYNCHRONIZED Clause

COMP SYNC (1 TO 4 Digits)	= 2 Byte boundary
COMP SYNC (5 TO 9 Digits)	= 4 Byte boundary
COMP SYNC (10 TO 18 Digits)	= 8 Byte boundary
INDEX	= 4 Byte boundary

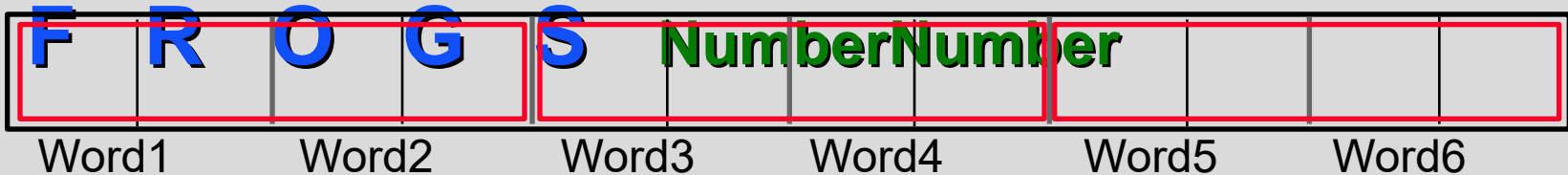
```
01 ThreeBytes  PIC XXX VALUE "DOG".  
01 TwoBytes   PIC 9(4) COMP SYNC.
```



The SYNCHRONIZED Clause

COMP SYNC (1 TO 4 Digits)	= 2 Byte boundary
COMP SYNC (5 TO 9 Digits)	= 4 Byte boundary
COMP SYNC (10 TO 18 Digits)	= 8 Byte boundary
INDEX	= 4 Byte boundary

```
01 FiveBytes    PIC XXX VALUE "FROGS".  
01 FourBytes    PIC S9(5)  COMP.
```



The SYNCHRONIZED Clause

COMP SYNC (1 TO 4 Digits)	= 2 Byte boundary
COMP SYNC (5 TO 9 Digits)	= 4 Byte boundary
COMP SYNC (10 TO 18 Digits)	= 8 Byte boundary
INDEX	= 4 Byte boundary

```
01 FiveBytes    PIC XXX VALUE "FROGS".  
01 FourBytes    PIC S9(5)  COMP SYNC.
```

Aligns along a
4 byte boundary

